

Sequential Monitoring of Common-Factor Structural Breaks in Heterogeneous Panels: A Brownian-Bridge CUSUM-of-Squares Procedure — Design, Closed-Form Reductions, and Computational Implementation

Abstract

This note documents, in full mathematical and econometric detail, a Monte Carlo design for evaluating the finite-sample size and power of a sequential structural-break monitoring procedure applied to a cross-sectionally heterogeneous panel subject to a common-factor shock. The procedure pre-filters each unit with a fixed-coefficient autoregression estimated on a calibration sample, aggregates the resulting residuals across units via a pooled cross-pair regression, and monitors the cumulative sum of squares of the aggregated residual sequence against a boundary derived from the Brownian-bridge limiting distribution. We give (i) the exact data generating processes (heterogeneous ARMA/ARFIMA panels and common factors), (ii) a closed-form algebraic reduction of the $N(N - 1)$ -pair pooled-OLS aggregation step to a single cross-sectional average, (iii) the sequential decision rule and the associated stopping time, (iv) the two-scenario Monte Carlo design (empirical size under the null of no break, and empirical power / detection-delay under a known break location), and (v) the computational reformulations — an FFT-based fractional-differencing filter and a vectorized AR-residual recursion — that reduce the per-replication cost by roughly two orders of magnitude without altering any statistic.

1 The Monitoring Problem

Consider a panel of N time series $\{Y_{i,t}\}_{i=1,\dots,N; t=1,\dots,T}$ that are individually heterogeneous in their short- and long-memory dynamics but share exposure to a common latent factor f_t . A practitioner observes the panel up to the end of a calibration window of length M ($t = 1, \dots, M$), during which the panel is assumed to be free of structural change, and then monitors the panel sequentially for $t = M + 1, M + 2, \dots$ for evidence of a common-mean shift.

The null hypothesis of interest is

$$H_0 : \text{no structural change occurs in } \{Y_{i,t}\}_{i=1}^N \text{ for any } t > M,$$

against the alternative

$$H_1 : \text{at some calendar time } \tau > M, \text{ all } N \text{ series receive a common level shift of size } \delta \neq 0.$$

The monitoring procedure consists of four stages, detailed in Sections 2–5: (a) generation of the heterogeneous panel and the common-factor-augmented observables; (b) unit-by-unit fixed-coefficient AR pre-filtering, calibrated on $t = 1, \dots, M$; (c) cross-sectional aggregation of the filtered residuals into a single scalar sequence $\{\bar{e}_t\}$ via a pooled cross-pair regression; and (d) sequential monitoring of $\{\bar{e}_t\}_{t>M}$ via a CUSUM-of-squares statistic compared against a Brownian-bridge boundary. Section 7 then specifies the two-scenario Monte Carlo design used to assess the procedure’s empirical size and power, and Section 8 documents the computational reformulations used to make the 1.9×10^6 -replication design feasible.

2 Data Generating Processes

2.1 Heterogeneous panel

Each series $Y_{i,t}$ follows an ARFIMA(p_i, d_i, q_i) process,

$$\Phi_i(L)(1 - L)^{d_i}Y_{i,t} = \Theta_i(L)\varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \text{i.i.d. } N(0, 1),$$

where $\Phi_i(L) = 1 - \phi_i L$ and $\Theta_i(L) = 1 + \theta_i L$ are (at most) first-order AR and MA polynomials and $(1 - L)^{d_i}$ is the fractional differencing operator. Two cross-sectional designs are used.

Design 1 ($N = 10$). The ten units are grouped into five pairs of identical specifications, summarized in Table 1.

Table 1: Data-generating processes for $N = 10$.

Series i	Model	ϕ_i	d_i	θ_i
1, 2	ARMA(1, 1)	0.8	0	0.5
3, 4	ARMA(1, 1)	0.5	0	0.3
5, 6	ARFIMA(0, d , 0)	0	0.4	0
7, 8	ARFIMA(1, d , 1)	0.1	0.3	0.8
9, 10	ARFIMA(1, d , 0)	0.7	0.4	0

Design 2 ($N = 30$). The thirty units form a random-coefficient panel: within each of the four blocks below, the AR, fractional-integration, and MA parameters are drawn *independently across units and independently for every Monte Carlo replication* from the indicated uniform distributions, summarized in Table 2.

Table 2: Data-generating processes for $N = 30$ (random-coefficient panel).

Series i	Model	ϕ_i	d_i	θ_i
1–15	ARMA(1, 1)	$\phi_i \sim U(0, 0.5)$	0	0.6
16–20	ARMA(1, 1)	$\phi_i \sim U(0.7, 0.9)$	0	0.6
21–25	ARFIMA(0, d , 0)	0	$d_i \sim U(0, 0.5)$	0
26–30	ARFIMA(1, d , 1)	$\phi_i \sim U(0, 0.4)$	$d_i \sim U(0, 0.5)$	$\theta_i \sim U(0, 0.3)$

Re-drawing the random coefficients on every replication implies that the Monte Carlo average is taken *jointly* over the coefficient distribution and over the innovation sequences (Section 7 discusses the implications of this choice for the interpretation of the resulting size/power estimates).

2.2 Common factors

Four common-factor processes $f_t^{(k)}$, $k = 1, \dots, 4$, are generated as ARFIMA($p, d, 0$) processes with parameters given in Table 3; they span the i.i.d., near-unit-root short-memory, and two long-memory ($d = 0.2$ and $d = 0.45$) cases.

Table 3: Common-factor data generating processes.

Factor	Model	ϕ	d
factor1	i.i.d. $N(0, 1)$	0	0
factor2	AR(1)	0.99	0
factor3	ARFIMA(0, d , 0)	0	0.20
factor4	ARFIMA(0, d , 0)	0	0.45

2.3 Factor-augmented observables and break injection

Letting $f_t \equiv f_t^{(k)}$ denote the selected common factor, the observed panel is

$$Y_{i,t}^* = Y_{i,t} + \lambda f_t, \quad \lambda = 0.5, \quad i = 1, \dots, N, \quad (1)$$

i.e. a single-factor model with a common, unit-invariant factor loading $\lambda = 0.5$. Under the alternative hypothesis H_1 , a deterministic, common level shift of magnitude δ is additionally injected at a calendar time τ :

$$Y_{i,t}^{**} = Y_{i,t}^* + \delta \cdot \mathbf{1}\{t \geq \tau\}, \quad i = 1, \dots, N. \quad (2)$$

The construction of τ in terms of the calibration length M is given in Section 7.2.

3 Fixed-Coefficient AR Pre-Filtering

For each unit i , an autoregression of order

$$p = \max\left(3, 3 \left\lfloor \frac{M-1}{100} \right\rfloor\right) \quad (3)$$

is estimated by ordinary least squares using only the calibration sample $t = 1, \dots, M$:

$$\hat{\beta}_{AR} = \arg \min_{b \in \mathbb{R}^p} \sum_{t=p+1}^M \left(Y_{i,t} - \sum_{\ell=1}^p b_{\ell} Y_{i,t-\ell} \right)^2. \quad (4)$$

The estimated coefficient vector $\hat{\beta}_{AR}$ is then held *fixed* for the remainder of the sample, and out-of-calibration (one-step-ahead) residuals are computed as

$$\hat{e}_{i,t} = Y_{i,t} - \hat{\beta}_{AR}' (Y_{i,t-1}, \dots, Y_{i,t-p})', \quad t = p+1, \dots, T, \quad (5)$$

with the first seven observations additionally discarded for numerical burn-in (i.e. $\hat{e}_{i,t}$ is defined for $t \geq 8$ in the implementation, which is non-binding whenever $p < 8$, the case throughout this design since $p \leq 6$ for $M \leq 300$). The purpose of (5) is to remove each unit's idiosyncratic short- and long-range dependence *using only information available at the calibration date*, so that, under H_0 , the sequence $\{\hat{e}_{i,t}\}_{t \geq M}$ is approximately a mean-zero, covariance-stationary process to which the asymptotic theory of Section 5 applies. The vectorized (and exactly equivalent) evaluation of (5) is discussed in Section 8.2.

4 Cross-Sectional Aggregation: A Closed-Form Reduction

4.1 Pooled cross-pair regression

Let $\hat{e}_{i,t}$, $i = 1, \dots, N$, $t = 1, \dots, T$, denote the AR-filtered residuals from Section 3. Consider the set of all *ordered pairs* (i, j) with $i \neq j$, of which there are $N(N-1)$. For each ordered pair, postulate the calibration-sample linear relation

$$\hat{e}_{i,t} = \alpha + \beta \hat{e}_{j,t} + u_{i,j,t}, \quad t = 1, \dots, M. \quad (6)$$

A *single* pair of coefficients $(\hat{\alpha}, \hat{\beta})$ is estimated by ordinary least squares on the pooled sample of $N(N-1)M$ observations obtained by stacking (6) over all ordered pairs and $t = 1, \dots, M$. The full-sample (out-of-calibration included) residual for pair (i, j) is then

$$U_{i,j,t} = \hat{e}_{i,t} - \hat{\alpha} - \hat{\beta} \hat{e}_{j,t}, \quad t = 1, \dots, T. \quad (7)$$

The monitoring statistic of Section 5 is constructed from the cross-pair average

$$\bar{e}_t \equiv \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N U_{i,j,t}, \quad t = 1, \dots, T. \quad (8)$$

Naively, (8) requires forming and averaging $N(N-1)$ residual sequences, and the pooled OLS estimation in (6) requires a design matrix with $N(N-1)M$ rows. Propositions 1 and 2 below show that both steps admit closed forms of order $O(NM)$ and $O(NT)$, respectively, that are *exactly* (not approximately) equivalent to the pooled-pair construction.

4.2 Closed form for $(\hat{\alpha}, \hat{\beta})$

Define, for $t = 1, \dots, M$, the cross-sectional mean $\bar{\hat{e}}_t = \frac{1}{N} \sum_{i=1}^N \hat{e}_{i,t}$, the grand calibration mean

$$\mu \equiv \frac{1}{NM} \sum_{i=1}^N \sum_{t=1}^M \hat{e}_{i,t} = \frac{1}{M} \sum_{t=1}^M \bar{\hat{e}}_t, \quad (9)$$

and the demeaned individual residuals $d_{i,t} \equiv \hat{e}_{i,t} - \mu$ for $t \leq M$. Define the cross-sectional sum and sum-of-squares of the demeaned residuals,

$$D_t \equiv \sum_{i=1}^N d_{i,t}, \quad Q_t \equiv \sum_{i=1}^N d_{i,t}^2, \quad t = 1, \dots, M. \quad (10)$$

Proposition 1 (Closed-form pooled-OLS coefficients). *The pooled-OLS estimates from (6) satisfy*

$$\hat{\beta} = \frac{\sum_{t=1}^M (D_t^2 - Q_t)}{(N-1) \sum_{t=1}^M Q_t}, \quad \hat{\alpha} = \mu (1 - \hat{\beta}). \quad (11)$$

Proof. Write the pooled-OLS formulas for (6) as $\hat{\beta} = \text{Cov}_{\text{pool}} / \text{Var}_{\text{pool}}$ and $\hat{\alpha} = \bar{y}_{\text{pool}} - \hat{\beta} \bar{x}_{\text{pool}}$, where the pooled sample consists of $N(N-1)M$ triples $\{(\hat{e}_{i,t}, \hat{e}_{j,t})\}_{i \neq j, t \leq M}$ with $\hat{e}_{i,t}$ in the role of the dependent variable and $\hat{e}_{j,t}$ in the role of the regressor.

Pooled means coincide with μ . For fixed i , the number of ordered pairs in which i appears as the first index is $N-1$ (one for each $j \neq i$); hence

$$\bar{y}_{\text{pool}} = \frac{1}{N(N-1)M} \sum_{i \neq j} \sum_{t \leq M} \hat{e}_{i,t} = \frac{1}{N(N-1)M} (N-1) \sum_{i=1}^N \sum_{t \leq M} \hat{e}_{i,t} = \frac{1}{NM} \sum_{i,t \leq M} \hat{e}_{i,t} = \mu,$$

and by the identical argument with the roles of i and j exchanged, $\bar{x}_{\text{pool}} = \mu$ as well.

Numerator. The (unnormalized) pooled covariance is

$$\sum_{i \neq j} \sum_{t \leq M} (\hat{e}_{i,t} - \mu)(\hat{e}_{j,t} - \mu) = \sum_{t=1}^M \sum_{i \neq j} d_{i,t} d_{j,t}.$$

For fixed t , expand the square of the cross-sectional sum:

$$\left(\sum_{i=1}^N d_{i,t} \right)^2 = \sum_{i=1}^N d_{i,t}^2 + \sum_{i \neq j} d_{i,t} d_{j,t} \implies \sum_{i \neq j} d_{i,t} d_{j,t} = D_t^2 - Q_t.$$

Summing over $t = 1, \dots, M$ gives $\sum_{t=1}^M (D_t^2 - Q_t)$.

Denominator. The (unnormalized) pooled variance of the regressor is

$$\sum_{i \neq j} \sum_{t \leq M} (\hat{e}_{j,t} - \mu)^2 = \sum_{t=1}^M \sum_{j=1}^N \underbrace{\#\{i \neq j\}}_{=N-1} d_{j,t}^2 = (N-1) \sum_{t=1}^M Q_t.$$

Taking the ratio of the numerator and denominator gives the expression for $\hat{\beta}$ in (11); substituting $\bar{y}_{\text{pool}} = \bar{x}_{\text{pool}} = \mu$ into $\hat{\alpha} = \bar{y}_{\text{pool}} - \hat{\beta} \bar{x}_{\text{pool}}$ gives $\hat{\alpha} = \mu(1 - \hat{\beta})$. $\square$

Proposition 1 reduces the estimation of $(\hat{\alpha}, \hat{\beta})$ from an $O(N^2M)$ regression on a stacked $N(N-1)M \times 2$ design matrix to $O(NM)$ arithmetic operations on the M -vectors D_t and Q_t .

4.3 Closed form for the cross-pair average $\bar{e}_t$

Proposition 2 (Closed-form cross-pair average). *For every $t = 1, \dots, T$,*

$$\bar{e}_t = (1 - \hat{\beta})(\bar{\hat{e}}_t - \mu), \quad \bar{\hat{e}}_t \equiv \frac{1}{N} \sum_{i=1}^N \hat{e}_{i,t}. \quad (12)$$

Proof. Let $S_t \equiv \sum_{i=1}^N \hat{e}_{i,t} = N\bar{\hat{e}}_t$. From the definition (8) and (7),

$$\bar{e}_t = \frac{1}{N(N-1)} \sum_{j=1}^N \sum_{i \neq j} (\hat{e}_{i,t} - \hat{\alpha} - \hat{\beta} \hat{e}_{j,t}).$$

Split the sum into three parts.

First term. For fixed j , $\sum_{i \neq j} \hat{e}_{i,t} = S_t - \hat{e}_{j,t}$. Summing over $j = 1, \dots, N$,

$$\sum_{j=1}^N (S_t - \hat{e}_{j,t}) = NS_t - S_t = (N-1)S_t.$$

Second term. $\sum_{j=1}^N \sum_{i \neq j} \hat{\alpha} = \sum_{j=1}^N (N-1)\hat{\alpha} = N(N-1)\hat{\alpha}$.

Third term. $\sum_{j=1}^N \sum_{i \neq j} \hat{\beta} \hat{e}_{j,t} = \hat{\beta} \sum_{j=1}^N (N-1)\hat{e}_{j,t} = (N-1)\hat{\beta} S_t$.

Collecting terms,

$$\bar{e}_t = \frac{(N-1)S_t - N(N-1)\hat{\alpha} - (N-1)\hat{\beta} S_t}{N(N-1)} = \frac{(1-\hat{\beta})S_t}{N} - \hat{\alpha} = (1-\hat{\beta})\bar{\hat{e}}_t - \hat{\alpha}.$$

Substituting $\hat{\alpha} = \mu(1 - \hat{\beta})$ from Proposition 1 yields $\bar{e}_t = (1 - \hat{\beta})\bar{\hat{e}}_t - \mu(1 - \hat{\beta}) = (1 - \hat{\beta})(\bar{\hat{e}}_t - \mu)$. $\square$

Remark 1. *Proposition 2 shows that, because every one of the $N(N - 1)$ ordered-pair regressions shares the same pooled coefficients $(\hat{\alpha}, \hat{\beta})$, averaging the $N(N - 1)$ pair-level residuals $U_{i,j,t}$ is algebraically identical to (i) computing the simple cross-sectional mean $\bar{\tilde{e}}_t$ of the AR-filtered residuals at every t , and (ii) applying a single affine transformation $x \mapsto (1 - \hat{\beta})(x - \mu)$ with coefficients estimated once from (11). The full pipeline from $\{\hat{e}_{i,t}\}$ to $\{\bar{\tilde{e}}_t\}$ therefore costs $O(NM)$ for $(\hat{\alpha}, \hat{\beta}, \mu)$ plus $O(NT)$ for $\bar{\tilde{e}}_t$, i.e. $O(NT)$ overall, as opposed to $O(N^2M)$ for the naive pairwise construction — a reduction by a factor of $O(N)$, which is material for $N = 30$ ($N(N - 1) = 870$ pairs).*

5 The CUSUM-of-Squares Statistic and the Brownian-Bridge Boundary

Given the scalar sequence $\{\bar{\tilde{e}}_t\}_{t=1}^T$ from (12), define the cumulative sum of squares

$$C_n \equiv \sum_{t=1}^n \bar{\tilde{e}}_t^2, \quad n = 1, \dots, T, \quad (13)$$

and, for monitoring times $n = M + 1, \dots, T$, the standardized statistic

$$T_n \equiv \sqrt{\frac{M}{2}} \left(\frac{C_n}{C_M} - \frac{n-2}{M-2} \right). \quad (14)$$

Under H_0 and standard regularity conditions on $\{\bar{\tilde{e}}_t\}$ (the filtering and aggregation steps of Sections 3–4 are designed precisely to make these conditions plausible), the process $\{T_{\lfloor M+s(T-M) \rfloor}\}_{s \in [0,1]}$, when reparametrized in terms of the ratio

$$r \equiv \frac{n-M}{M}, \quad q \equiv \frac{n}{n-M} = 1 + \frac{1}{r}, \quad (15)$$

converges to a functional of a standard Brownian bridge $W^0(\cdot)$ on $[0, 1]$, $W^0(0) = W^0(1) = 0$. The associated asymptotic boundary-crossing probabilities for $\sup_s |W^0(s)|$ at nominal levels $\alpha \in \{10\%, 5\%, 1\%\}$ correspond to the constants

$$a_{10\%}^2 = 6.2514, \quad a_{5\%}^2 = 7.8147, \quad a_{1\%}^2 = 11.3449, \quad (16)$$

which are the upper- α quantiles of a χ_3^2 distribution. The corresponding finite-sample boundary function, expressed in (r, q) from (15), is

$$g(n; a_\alpha^2) = r \sqrt{q(a_\alpha^2 + \ln q)}. \quad (17)$$

6 Sequential Decision Rule and Stopping Time

The monitoring procedure rejects H_0 at level α at the first monitoring time at which $|T_n|$ exits the boundary (17). Formally, define the *stopping time*

$$\hat{\tau}_\alpha \equiv \inf\{n > M : |T_n| \geq g(n; a_\alpha^2)\}, \quad \hat{\tau}_\alpha = +\infty \text{ if no such } n \leq T \text{ exists.} \quad (18)$$

All quantities reported in Section 7 are functions of $\hat{\tau}_\alpha$ for $\alpha \in \{10\%, 5\%, 1\%\}$, evaluated on a per-replication basis.

7 Monte Carlo Design

Two scenarios are considered, corresponding to the two states of nature H_0 and H_1 defined in Section 1. Both scenarios share the sample-length convention

$$T_{\text{total}} = 10M + 10, \quad (19)$$

i.e. the monitoring window extends to nine calibration-lengths beyond M , which is ample for all break locations and scoring windows considered below ($k \leq 3$).

7.1 Scenario A: empirical size ($\delta = 0$)

Under Scenario A no break is injected ($\delta = 0$ in (2)), so H_0 holds throughout the monitoring window for every replication. Because the data-generating process does not depend on k , a *single* simulated path $\{\bar{e}_t\}_{t=1}^{T_{\text{total}}}$ — and hence a single stopping time $\hat{\tau}_\alpha^{(r)}$ from (18) — can be scored against $K = 6$ post-hoc evaluation windows of length kM , $k \in \mathcal{K} \equiv \{0.2, 0.3, 0.5, 1, 2, 3\}$. Define, for replication $r = 1, \dots, R$,

$$A_\alpha^{(r)}(k) \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq M + kM\}, \quad k \in \mathcal{K}, \alpha \in \{10\%, 5\%, 1\%\}. \quad (20)$$

The empirical size for window k is the Monte Carlo average

$$\widehat{\text{Size}}_\alpha(N, m, \text{factor}, k) = \frac{1}{R} \sum_{r=1}^R A_\alpha^{(r)}(k), \quad (21)$$

which estimates $\Pr_{H_0}(\hat{\tau}_\alpha \leq M + kM)$, the probability of a false alarm occurring within the first kM post-calibration observations. A well-calibrated procedure should exhibit $\widehat{\text{Size}}_\alpha \approx \alpha$ for k large enough that the boundary has stabilized, with $\widehat{\text{Size}}_\alpha$ monotonically non-decreasing in k (a longer monitoring horizon affords more opportunities for a false alarm).

7.2 Scenario B: empirical power, premature alarms, and detection delay ($\delta \in \{1, 2, 3\}$)

Under Scenario B, H_1 holds: a break of magnitude $\delta \in \{1, 2, 3\}$ is injected at the calendar time

$$\tau(k) \equiv M + \lfloor kM \rfloor, \quad k \in \mathcal{K} = \{0.2, 0.3, 0.5, 1, 2, 3\}, \quad (22)$$

i.e. k now parametrizes the *break location* relative to the calibration length, rather than a post-hoc evaluation window as in Scenario A. Because the data-generating process depends on (δ, k) , each (δ, k) combination requires its own set of R independent replications.

For each replication, the stopping time $\hat{\tau}_\alpha^{(r)}$ from (18) is computed on the broken series, and is classified relative to the true break date $\tau(k)$ into three mutually exclusive outcomes:

$$P_\alpha^{(r)} \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq \tau(k)\} \quad (\text{premature alarm: triggered before the break occurs}), \quad (23)$$

$$D_\alpha^{(r)} \equiv \mathbf{1}\{\tau(k) < \hat{\tau}_\alpha^{(r)} < \infty\} \quad (\text{post-break detection}), \quad (24)$$

$$B_\alpha^{(r)} \equiv P_\alpha^{(r)} + D_\alpha^{(r)} = \mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\} \quad (\text{any alarm}). \quad (25)$$

The Monte Carlo averages

$$\hat{P}_\alpha = \frac{1}{R} \sum_{r=1}^R P_\alpha^{(r)}, \quad \hat{D}_\alpha = \frac{1}{R} \sum_{r=1}^R D_\alpha^{(r)}, \quad \hat{B}_\alpha = \hat{P}_\alpha + \hat{D}_\alpha \quad (26)$$

estimate, respectively, the probability of a false alarm *before* the break, the (strict) power of the procedure — the probability of correctly flagging the break *after* it occurs — and the overall alarm probability.

For replications with $D_\alpha^{(r)} = 1$, the *detection delay*

$$\text{Delay}_\alpha^{(r)} \equiv \hat{\tau}_\alpha^{(r)} - \tau(k) > 0 \quad (27)$$

is recorded, and its empirical distribution over the subset $\{r : D_\alpha^{(r)} = 1\}$ is summarized by the sample mean, standard deviation, median, and the 25th/75th percentiles. Conditioning the delay distribution on $D_\alpha^{(r)} = 1$ avoids mixing two conceptually distinct events (a premature, pre-break alarm with $\hat{\tau}_\alpha^{(r)} - \tau(k) < 0$, and a genuine post-break detection delay) into a single statistic.

7.3 Full nested design

The complete experiment is a four-way (Scenario A) or five-way (Scenario B) factorial design, summarized in Table 4. For every admissible parameter combination, $R = 2500$ independent replications are drawn; for Design 2 ($N = 30$), the random AR/fractional-integration/MA coefficients of Table 2 are redrawn independently on *every* replication, so that $\widehat{\text{Size}}_\alpha$, $\hat{P}_\alpha$, $\hat{D}_\alpha$ etc. estimate probabilities taken jointly over the coefficient distribution *and* the innovation sequences — an average performance measure over the class of data-generating processes described in Table 2, rather than a measure conditional on one particular coefficient draw.

8 Computational Implementation

The full design of Table 4 requires 1.9×10^6 independent replications. A direct (unoptimized) implementation of the fractional-differencing filter and the AR-residual recursion — both present in every replication — has complexity that renders this infeasible; this section derives two reformulations that are *numerically exact* (i.e. produce the same statistics to machine precision) but reduce the asymptotic and constant-factor cost substantially.

Table 4: Nested Monte Carlo design.

Factor	Scenario A (size)	Scenario B (power)
Cross-sectional dimension N	$\{10, 30\}$	$\{10, 30\}$
Break magnitude δ	— ($\delta = 0$)	$\{1, 2, 3\}$
Index k	post-hoc scoring window $M + kM$ $\{0.2, 0.3, 0.5, 1, 2, 3\}$	break location $\tau(k) = M$ $\{0.2, 0.3, 0.5, 1, 2, 3\}$
Calibration length M	$\{50, 80, 100, 200, 300\}$	$\{50, 80, 100, 200, 300\}$
Common factor	$\{1, 2, 3, 4\}$	$\{1, 2, 3, 4\}$
Replications R	2500	2500
Distinct (N, M, factor) combinations	$2 \times 5 \times 4 = 40$	—
Distinct $(N, \delta, k, M, \text{factor})$ combinations	—	$2 \times 3 \times 6 \times 5 \times 4 = 720$
Total replications	$40 \times 2500 = 10^5$	$720 \times 2500 = 1.8 \times 10^6$

8.1 FFT-based fractional differencing

An ARFIMA(p, d, q) process admits the representation $X_t = (1 - L)^{-d} \Psi(L) \varepsilon_t$, where $\Psi(L) = \Theta(L)/\Phi(L)$ is the (rational, short-memory) Wold filter and $(1 - L)^{-d}$ is the fractional integration operator. The binomial-series expansion of the latter is

$$(1 - L)^{-d} = \sum_{j=0}^{\infty} w_j L^j, \quad w_j = \frac{\Gamma(j + d)}{\Gamma(j + 1) \Gamma(d)}, \quad j = 0, 1, 2, \dots \quad (28)$$

Let $Y_t \equiv \Psi(L) \varepsilon_t$ be the (short-memory) intermediate series of length T , generated by a standard linear recursion (e.g. via `scipy.signal.lfilter` or an equivalent ARMA recursion, $O(T)$ cost). The finite-sample fractionally integrated series is the truncated causal convolution

$$X_t = \sum_{j=0}^t w_j Y_{t-j}, \quad t = 0, \dots, T - 1. \quad (29)$$

Equation (29) is precisely the first T entries of the full discrete convolution $(w * Y)$, where $w = (w_0, \dots, w_{T-1})$: by definition, $(w * Y)_t = \sum_m w_m Y_{t-m}$, and for $t < T$ only the terms $0 \leq m \leq t$ are non-zero (since $Y_{t-m} = 0$ for $t - m < 0$ in the finite array), reproducing (29) exactly.

A direct evaluation of (29) for $t = 0, \dots, T - 1$ costs $O(T^2)$ scalar multiply-adds. By the convolution theorem, however,

$$w * Y = \mathcal{F}^{-1}[\mathcal{F}(w) \odot \mathcal{F}(Y)], \quad (30)$$

where $\mathcal{F}$ denotes the discrete Fourier transform (computed via the Fast Fourier Transform) and $\odot$ is elementwise multiplication; both $\mathcal{F}$ and $\mathcal{F}^{-1}$ cost $O(T \log T)$, so the entire convolution (30), and hence (29), costs $O(T \log T)$. For $T = 4000$, this is the `scipy.signal.fftconvolve(Y, w)[:T]` operation, which we verified reproduces the direct $O(T^2)$ computation to floating-point precision.

For numerical stability, the weights (28) are computed in log-space to avoid overflow of $\Gamma(j + d)$ for j up to several thousand:

$$w_j = \exp\left[\ln \Gamma(j + d) - \ln \Gamma(j + 1) - \ln \Gamma(d)\right], \quad (31)$$

evaluated via the (vectorized) log-gamma function, which is itself $O(T)$.

Remark 2. *The combination of (31) (vectorized, $O(T)$) and (30) ($O(T \log T)$) replaces an $O(T^2)$ Python-level loop — which additionally incurred T separate interpreter-level calls to a NumPy reduction — with two vectorized array operations. Empirically, for $T = 4000$, this reduced the fractional-differencing step from on the order of hundreds of milliseconds to under one millisecond per series.*

8.2 Vectorized AR-residual recursion

Equation (5) is, as written, a recursion evaluated sequentially for $t = p + 1, \dots, T$ (subject to the $t \geq 8$ burn-in convention), each step requiring an inner product of two length- p vectors. A literal transcription — a **for** loop over t , each iteration invoking a NumPy dot product — incurs $T - p$ separate Python/NumPy call overheads c , for a wall-clock cost of order $(T - p)(c + O(p))$, with $c \gg O(p)$ for the small orders $p \in \{3, 6\}$ used here (p from (3)).

Lemma 1 (Vectorization equivalence). *Let $t_0 \equiv \max(8, p + 1)$. Define the index set $\mathcal{I} = \{t_0, \dots, T\}$ and the lag matrix $Z \in \mathbb{R}^{|\mathcal{I}| \times p}$ with rows*

$$Z_{t,\ell} = Y_{i,t-\ell}, \quad \ell = 1, \dots, p, \quad t \in \mathcal{I}. \quad (32)$$

Then the vector $\hat{e}_i = (\hat{e}_{i,t})_{t \in \mathcal{I}}$ defined by (5) satisfies

$$\hat{e}_i = y - Z\hat{\beta}_{AR}, \quad y \equiv (Y_{i,t})_{t \in \mathcal{I}}, \quad (33)$$

i.e. (5) for all $t \in \mathcal{I}$ is computed by a single matrix-vector product.

Proof. By construction (5) states $\hat{e}_{i,t} = Y_{i,t} - \sum_{\ell=1}^p \hat{\beta}_{AR,\ell} Y_{i,t-\ell}$ for each $t \in \mathcal{I}$. The t -th row of $Z\hat{\beta}_{AR}$ is, by (32), $\sum_{\ell=1}^p Z_{t,\ell} \hat{\beta}_{AR,\ell} = \sum_{\ell=1}^p \hat{\beta}_{AR,\ell} Y_{i,t-\ell}$, which is exactly the subtracted term. Hence $(y - Z\hat{\beta}_{AR})_t = \hat{e}_{i,t}$ for every $t \in \mathcal{I}$, establishing (33) entrywise, and therefore as a vector identity. $\square$

Lemma 1 is the AR-recursion analogue of the calibration-step design matrix already used in (4): both are instances of "linear recursion as a single linear-algebra operation." The right-hand side of (33) is evaluated as one BLAS-level matrix-vector product of size $|\mathcal{I}| \times p$, i.e. $O(|\mathcal{I}|p)$ floating-point operations performed in a single call, eliminating the $|\mathcal{I}|$ -fold interpreter overhead c of the loop formulation. We verified, for every $M \in \{50, 80, 100, 200, 300\}$ (and the corresponding $p \in \{3, 6\}$ from (3)), that (33) reproduces the sequential recursion (5) to within floating-point tolerance for all $t \in \mathcal{I}$. Empirically, this reduced the per-series AR-filtering cost by a factor of approximately 30 for $T \approx 3000$, $p \leq 6$.

8.3 Aggregate effect and parallelization

Combining Sections 8.1 and 8.2 with the $O(NT)$ closed-form aggregation of Section 4, the per-replication cost for the heaviest configuration ($N = 30$, $M = 300$) was reduced from roughly 1.4×10^{-1} seconds to roughly 1.8×10^{-2} seconds, i.e. by a factor of approximately 8; the dominant remaining cost is the $O(N)$ generation of N independent ARMA/ARFIMA series per replication. Across the full design of Table 4 (1.9×10^6 replications), this corresponds to a reduction in projected single-threaded runtime from approximately 72 hours to approximately 9.5 hours.

The 760 distinct parameter combinations of Table 4 are embarrassingly parallel: each combination's $R = 2500$ replications are independent both of one another and across combinations. The implementation therefore distributes combinations across 8 worker processes (one combination = 2500 replications = one task), with each worker process seeded independently (via OS entropy) to ensure that the pseudo-random streams used across processes do not overlap — a necessary condition for the Monte Carlo averages (21) and (26) to retain their interpretation as estimates of probabilities under 1.9×10^6 *independent* draws. With 8-way parallelism, the projected wall-clock runtime for the full design is on the order of 1–1.5 hours.

9 Summary of Reported Quantities

For each of the 40 Scenario-A combinations (N, M, factor) and each $k \in \mathcal{K}$, the design reports $\widehat{\text{Size}}_\alpha(N, M, \text{factor}, k)$ for $\alpha \in \{10\%, 5\%, 1\%\}$ (240 rows, Eq. 21). For each of the 720 Scenario-B combinations ($N, \delta, k, M, \text{factor}$), the design reports $\widehat{B}_\alpha, \widehat{P}_\alpha, \widehat{D}_\alpha$ and the mean, standard deviation, median, and 25th/75th percentiles of Delay_α conditional on $D_\alpha^{(r)} = 1$, for each $\alpha \in \{10\%, 5\%, 1\%\}$ (720 rows, Eqs. 26–27). Together, Propositions 1 and 2 guarantee that every one of these 1.9×10^6 replications is evaluated using the *exact* pooled cross-pair statistic of Section 4, despite the $O(N)$ (rather than $O(N^2)$) computational cost; and Section 8.1–8.2 guarantee that every replication's AR-filtered residuals and fractionally integrated components are numerically identical to those of a direct, unoptimized implementation.